Soumya Rani Samineni

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Summary

Third-year PhD student at Arizona State University (Yochan Lab) researching **reinforcement learning post-training, reasoning and evaluation of LLMs**. My findings show that **RLVR acts as a classification-based policy iteration method for learning to search**: GRPO updates reduce to a hinge loss, and the optimal policy learns a classification boundary on the base model's support, reinforcing valid and invalid reasoning alike.

- Ongoing work explores multi-turn and agentic RL for post-training, and evaluation of reasoning in non-ergodic worlds.
- Publications at ICML'26, NeurIPS'25 workshops and ICRA'22, a US patent application, and 2+ years of industry RL/ML experience at Quantiphi and Microsoft Research India.

Education

Arizona State University, Tempe, AZ — Aug 2024 – May 2029 (expected)
Ph.D., Computer Science and Engineering. Yochan Lab, advisor Prof. Subbarao Kambhampati — GPA: **4.0**/4.0

Indian Institute of Science, Bangalore — 2019 – 2021
M.Tech., Computer Science and Engineering. Thesis A+. Advisors: Profs. S. Bhatnagar, S. Kolathaya — CGPA: **8.5**/10

National Institute of Technology, Warangal. *B.Tech., Civil Engineering* — 2012 – 2016 CGPA: **8.55**/10

Research Projects

GRPO-Unhinge: The Hidden Advantage of Classification Updates in RLVR — *Ongoing*

- Proved that with binary rewards the clipped **GRPO update is a weighted hinge loss** on outcome-labeled tokens, with the score function as the decision boundary. This places RLVR in classification-based policy iteration and shows that GRPO variants (DAPO, GSPO, CISPO) differ only in how they set four components: class weights, margins, clipping and the score function.
- Proposed **GRPO-Unhinge** with per-class loss normalization. On 30×30 maze navigation it raised plan accuracy by **10% over GRPO** and kept incorrect responses at their original length, where GRPO **doubled** them.

RLVR as Prompt Augmentation Reinforces Both Correct and Incorrect Reasoning to the Correct Answer — *In review, ICLR'27*

- Derived the optimal policy of the RLVR objective as the **base policy conditioned on success**, and proved that a chain of thought is up-weighted only by how much it raises the solve probability, regardless of its validity or coherence.
- Showed on math reasoning benchmarks that GRPO raises accuracy while the rate of correct answers reached through *incoherent* reasoning also rises in **every setting**, contradicting the claim that RLVR implicitly incentivizes correct reasoning.

RL in Name Only? Structural Assumptions in RL Post-Training for LLMs — *NeurIPS'25 LAW workshop*

- Showed that, under the LLM-as-MDP assumptions used in RLVR, **on-policy GRPO reduces to filtered iterative SFT**; confirmed it empirically by post-training Qwen2.5, Llama 3.2 and DeepSeek 7B with GRPO and iterative SFT on GSM8K.
- Traced the growth in response length under RLVR to these structural assumptions rather than to emergent reasoning.

Local Coherence or Global Validity? Investigating RLVR Traces in Math — *NeurIPS'25 MATH-AI workshop*

- Built an LLM-as-judge pipeline that translates reasoning traces into first-order logic and tags errors, validated against human annotations.
- Showed that RLVR improves **local step coherence without making solutions valid**.

Inference-Time Scaling with Rollouts, with Prof. Dimitri Bertsekas — *Jan – May 2025*

- Applied rollouts at every decoding step, with an LLM as base policy and a verifier, on Game of 24, as an alternative to Tree of Thoughts.

Earlier PhD Projects — *2024 – 2025*

- VLM-generated rewards for inverse RL (GPT-4V, CLIP): evaluated whether VLM rewards can discriminate a learned policy's behavior from expert demonstrations.
- Multi-agent LLM framework for logic-grid puzzles, in which agents iteratively process clues and refine solutions.

Experience

Graduate Researcher, Yochan Lab, School of Computing and Augmented Intelligence, ASU — Aug 2024 – Present

- Theory and experiments on RL post-training of LLMs, reasoning and planning, and evaluation of reasoning traces (projects above).

Machine Learning Research Engineer, Quantiphi Analytics, India — Jul 2023 – Aug 2024

- Built an RL scheduler for flexible job-shop manufacturing with graph attention networks (TensorFlow, GCP); co-inventor on US patent application US20240319718A1.

Research Fellow, Microsoft Research India — Feb 2022 – Jul 2022

- Built the RL framework for a renewable-energy decision platform that handles carbon arbitrage, demand matching and bidding.

Solution Leader, Brane Enterprises, India Jul 2021 – Feb 2022

- Built a quadruped locomotion controller in lightweight C++ for microcontrollers (CAN, UART, ROS), cutting control latency to microseconds from the milliseconds reported in the source paper.

Graduate Researcher, Stochastic Systems and Stochastic Robotics Labs, IISc Bangalore Mar 2020 – Jul 2021

- Developed DeMo RL, a model-based lookahead for off-policy RL: **30% better sample efficiency** in simulation and a **2× gain** on a real 2R arm and quadruped (ICRA'22; NeurIPS'21 Deep RL and Offline RL workshops).

Assistant Executive Engineer, Roads and Buildings Department, Government of Telangana Jun 2018 – Jul 2019

Publications and Patents

- *Under review*. RLVR as Prompt Augmentation Reinforces Both Correct and Incorrect Reasoning to the Correct Answer. ICLR'27.
- Kambhampati S., Valmeekam K., Bhambri S., Palod V., Saldyt L., Stechly K., **Samineni S.R.**, Kalwar D., Biswas U. Position: Stop Anthropomorphizing Intermediate Tokens as Reasoning/Thinking Traces! ICML'26.
- **Samineni S.R.**, Kalwar D., Valmeekam K., Stechly K., Kambhampati S. RL in Name Only? Analyzing the Structural Assumptions in RL Post-Training for LLMs. NeurIPS'25 Workshop on Bridging Language, Agent, and World Models (LAW).
- **Samineni S.R.**, Kalwar D., Gangal V., Bhambri S., Kambhampati S. Local Coherence or Global Validity? Investigating RLVR Traces in Math Domains. NeurIPS'25 Workshop on Mathematical Reasoning and AI.
- **Samineni S.R.***, Mishra U.*, Goel P., Kunjeti C., Lodha H., Singh A., Sagi A., Bhatnagar S., Kolathaya S. Dynamic Mirror Descent based Model Predictive Control for Accelerating Robot Learning. ICRA'22. (*equal contribution)
- Birru D., Deodhar A., Chaudhary A., **Samineni S.R.** System and Method for Intelligent Scheduling of Manufacturing Jobs. US Patent Application US20240319718A1, 2024.

Technical Skills

Languages: Python, C++
LLM and RL: PyTorch, verl, vLLM, Ray, Hugging Face Transformers, TensorFlow, RLlib, Stable Baselines, MuJoCo
Systems: AWS, GCP, Azure, Docker, ROS

Honors and Certifications

- Student Research Grant, Robert Bosch Centre for Cyber Physical Systems, IISc (2021)
- GATE Computer Science and Engineering 2018: rank 136 of 107,893
- IIT JEE 2012: All-India Rank 6,062 (98.67 percentile)
- Applied Data Science with Python Specialization, University of Michigan (2023)
- TensorFlow Developer Certificate, DeepLearning.AI (2021)
- Winter School on Hybrid Cloud, IBM Research India and IISc Bangalore (2020)
- Summer School on Machine Learning and Computer Vision, IIIT Hyderabad (2019)